

Rex Ouyang

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EDUCATION

University of California, Berkeley

May. 2028

B.A. Computer Science, B.A. Applied Mathematics; EECS Honors Program

Berkeley, CA

- **Coursework:** Nonstochastic Statistics*, Stochastic Estimation and Control*, Applications of Stochastic Process Theory*, Random Processes in Systems*, Convex Optimization*, Topology and Analysis*, Deep Learning

* denotes graduate coursework

SELECTED PROJECTS & EXPERIENCE

Undergraduate Student Researcher, UC Berkeley EECS

May. 2026 – Present

Researching sharp entropy bounds in Euclidean spaces and geometricity conditions under Prof. Thomas Courtade

- Studying information-theoretic functional inequalities using Convex Analysis and Stochastic Calculus
- Analyzing reductions from entropy optimization over probability measures to finite-dimensional covariances

Undergraduate Course Staff, UC Berkeley EECS

Incoming Aug. 2026

1 out of 3 Selected Tutors for UC Berkeley EECS126: Probability and Random Processes, for Fall 2026 semester

- Will lead office hours, answer student questions, grade assignments and exams, and support administration

An Exposition of the Sharp Symmetrized Talagrand Inequality

May. 2026 – Present

Research project studying a [recent result](#) on optimal transport bounds measuring distance to centered distributions

- Using Convex Analysis and Stochastic Calculus, provided alternate proofs to key results in [this report](#)
- Building Monte Carlo simulations on probability distributions to test and visualize the inequality numerically

Entropy-Based Optimization for Wordle Solving

Mar. 2025 – Apr. 2025

Designed and evaluated an entropy-based search algorithm for the New York Times Wordle Puzzle to explore adaptive decision strategies in a constrained, information-limited environment

- Modeled the Wordle solving process as a sequential decision-making problem under uncertainty, using probabilistic belief updates and posterior filtering to adaptively reduce the search space after each guess
- Achieved an average solve rate of 3.283 guesses per game, outperforming the human average of 3.92 guesses

Simulations Developer, Formula Electric at Berkeley

Sep. 2024 – Present

Collaborating within a multidisciplinary engineering team to model battery discharge dynamics under varying load conditions, supporting performance analysis and system-level decision-making

- Designed a mathematical framework to estimate battery state under changing power demands using dynamical systems modeling and numerical simulation

Sparse Attack Detection in Power Networks via Lasso Optimization

Nov. 2025 – Dec. 2025

Formulated sensor attack detection as a sparse recovery problem using L1-regularized regression

- Evaluated sparse recovery performance under noisy and adversarial conditions by simulating corrupted measurement environments, enabling robust inference of hidden attack locations
- Achieved a 95% success rate in identifying all attack locations across repeated trials on simulated data

RISC-V Digit Classification

Jun. 2025 – Jul. 2025

Built a neural network in RISC-V assembly to classify handwritten digits from the MNIST dataset using matrix multiplication and file I/O routines.

- Implemented low-level numerical computation and data processing pipelines directly in RISC-V assembly
- Applied matrix-based computation techniques to support inference on digit classification

TECHNICAL SKILLS

- **Quantitative Methods:** Probability, Statistics, Stochastic Processes, Stochastic Analysis, Convex Optimization, Sparse Recovery, Statistical Modeling, Information Theory, Dynamical Systems, Machine Learning
- **Programming / Systems:** Java, Python, C/C++, SQL, Go, RISC-V Assembly, x86 Assembly
- **Scientific Computing / Tools:** NumPy, SciPy, CVXPY, Matplotlib, Git, GDB, Valgrind, Logisim, Unity